

# The Okinawan diet: health implications of a low-calorie, nutrient-dense, antioxidant-rich dietary pattern low in glycemic load.

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Residents of Okinawa, the southernmost prefecture of Japan, are known for their long average life expectancy, high numbers of centenarians, and accompanying low risk of age-associated diseases. Much of the longevity advantage in Okinawa is thought to be related to a healthy lifestyle, particularly the traditional diet, which is low in calories yet nutritionally dense, especially with regard to phytonutrients in the form of antioxidants and flavonoids. Research suggests that diets associated with a reduced risk of chronic diseases are similar to the traditional Okinawan diet, that is, vegetable and fruit heavy (therefore phytonutrient and antioxidant rich) but reduced in meat, refined grains, saturated fat, sugar, salt, and full-fat dairy products. Many of the characteristics of the diet in Okinawa are shared with other healthy dietary patterns, such as the traditional Mediterranean diet or the modern DASH (Dietary Approaches to Stop Hypertension) diet. Features such as the low levels of saturated fat, high antioxidant intake, and low glycemic load in these diets are likely contributing to a decreased risk for cardiovascular disease, some cancers, and other chronic diseases through multiple mechanisms, including reduced oxidative stress. A comparison of the nutrient profiles of the three dietary patterns shows that the traditional Okinawan diet is the lowest in fat intake, particularly in terms of saturated fat, and highest in carbohydrate intake, in keeping with the very high intake of antioxidant-rich yet calorie-poor orange-yellow root vegetables, such as sweet potatoes, and green leafy vegetables. Deeper analyses of the individual components of the Okinawan diet reveal that many of the traditional foods, herbs, or spices consumed on a regular basis could be labeled "functional foods" and, indeed, are currently being explored for their potential health-enhancing properties.